Results: replay of n = 83 rank block K1 and one timed K600 base construction

Verdict (frozen rule): `replay_pass`. Archived 2026-10-17 from GitHub Actions run 39007026856 (job 117321470019, conclusion success, head `d1c361e3`). Every gated check passed, on a host whose CPU model differs from the original run's, so the replay counts as another host class under the protocol's own definition. This report adds no new experiment and promotes, demotes or replaces no baseline row; it closes the stacked row's replication item (R1) for determinism and measures the construction time X that the row left uncharged (K2 / K-N1).

What was preregistered

PROTOCOL.md, frozen 2026-10-17 at commit `863f6090`, on its own, before the workflow commit `d1c361e3` (`.github/workflows/n83-replay.yml`) and before any run. The run's head is `d1c361e3`, so it executed the frozen protocol and the driver replay.py unchanged. Six runs in one job, one pinned CPU, one fresh process per run, in this order: R1 kS, R2 kP, R3 kP, R4 kS (rank block K1, point `s000`, rank seed 101, full rank, 600 attempts each, retained K600 base), R5 cS on Q0 loading the retained base (control), R6 cS on Q0 constructing the base in process (`construct:83:1:600`). The protocol fixed three pass conditions, the reported-not-gated numbers and the host-class rule before the job ran.

Evidence

run_github_39007026856/ is the workflow artifact `n83-replay-39007026856-1` (id 12094959671, 76,482 bytes, zip sha256 `b8226c794d7a3c5c7ab369487399f25869bff3661ca17c521c2c2da0b8fb951d`), downloaded through the GitHub API, checked against that digest, and unzipped byte for byte. ARTIFACT.json records its provenance and `SHA256SUMS` the digest of every file. The verdict and the numbers below are in panel/replay_result.json (identical to `analyze-stdout.json`); per-run records are in `panel/runs/0000{1..6}.stdout.txt` and `panel/runs.jsonl`.

Independent check (this archive, 2026-10-17). The per-attempt probe lists of R1-R4 were decoded from their `raw_b64` records and compared with the archived K1 records (`koblitz-n83-stacked-20261016/run_github_38862448935/panel/runs/00042` and `00043`, whose file digests equal the protocol's `d3bcbef8...be8d` and `2dd4bf0b...2396`): all 600 entries and the first 100 are equal for every run, and the sha256 of the compact JSON lists equals the protocol's digests (kS first 100 `8eed0160...64a6`, all `94856f30...791b`; kP first 100 `6081e1be...aafe8a`, all `d1d99f7e...cc6e`). `rank_relations_blake3` equals the archived value for every rank run. The pooled per-probe ratio was recomputed from the run rows (2.658328). Both IC and rho binaries have the same sha256 as in the original run (`ic_online_stacked_r770` `79c7bcd1...2e7b`, `rho_tuned_r770` `a426c244...0b2f`; `binaries.sha256` in both artifacts).

Host class

	original (run 38862448935)	replay (run 39007026856)
CPU model	AMD EPYC 9V74 80-Core Processor	INTEL(R) XEON(R) PLATINUM 8573C (family 6, model 207)
runner	GitHub `ubuntu-24.04`, 4 vCPU, Azure	GitHub `ubuntu-24.04` (image 20261012.334.1), 4 vCPU, 16 GB, Azure
kernel, compiler, Rust	6.17.0-1022-azure, gcc 13.3.0, rustc 1.99.0	the same

`another_host_class` is `true` by the frozen rule (CPU model differs). The VM hostname string (`runnervm8k3b6` in `host-probe.json`) is the same in both artifacts; it is set by the runner image, and the CPU vendor and model (`lscpu.txt`, `host-class.json`) show different hardware. Image, kernel and toolchain are the same, so the replay varies the CPU (vendor, microarchitecture, caches), not the software stack.

Gated criteria (frozen)

#	criterion	result
1	all six runs `valid` in `run_schedule.one()` / `arm.py` terms (exit 0, verified scalar, full rank, factor logs equal to B's table, index digests equal to B's)	**pass** (`all_runs_valid`; 6/6 `valid`, no issues)
2a	per-attempt probes equal the archived K1 records, first 100 attempts	**pass** (`probes_first100_equal_archived`)
2b	the same on all 600 attempts	**pass** (`probes_all_equal_archived`)
2c	`rank_relations_blake3` equals the archived value	**pass** (`relations_equal_archived`)
3a	R5 loads the retained base (`base_source retained_header`)	**pass** (`R5_retained`)
3b	R6 constructs it (`constructed_in_process_x_scan`, `base_scanned_x` 1266)	**pass** (`R6_constructed`)
3c	R6's base hash equals R5's (the retained base)	**pass** (`constructed_base_hash_equals_retained`; both `d4a4da26d6e7cb09367de922d9806631fe49989f70d378cae76f347f8d76a9f7`)

Verdict: **`replay_pass`** (`replay-exit.txt`: `replay_exit=0`).

The protocol abbreviates the base hash as `d4a4da26...f9e7`; the full hash used by every earlier n = 83 experiment, and observed here, ends in `...a9f7`. The abbreviation in the frozen text is a typo; criterion 3 compares R6 with R5, so it does not depend on it.

Reported, not gated

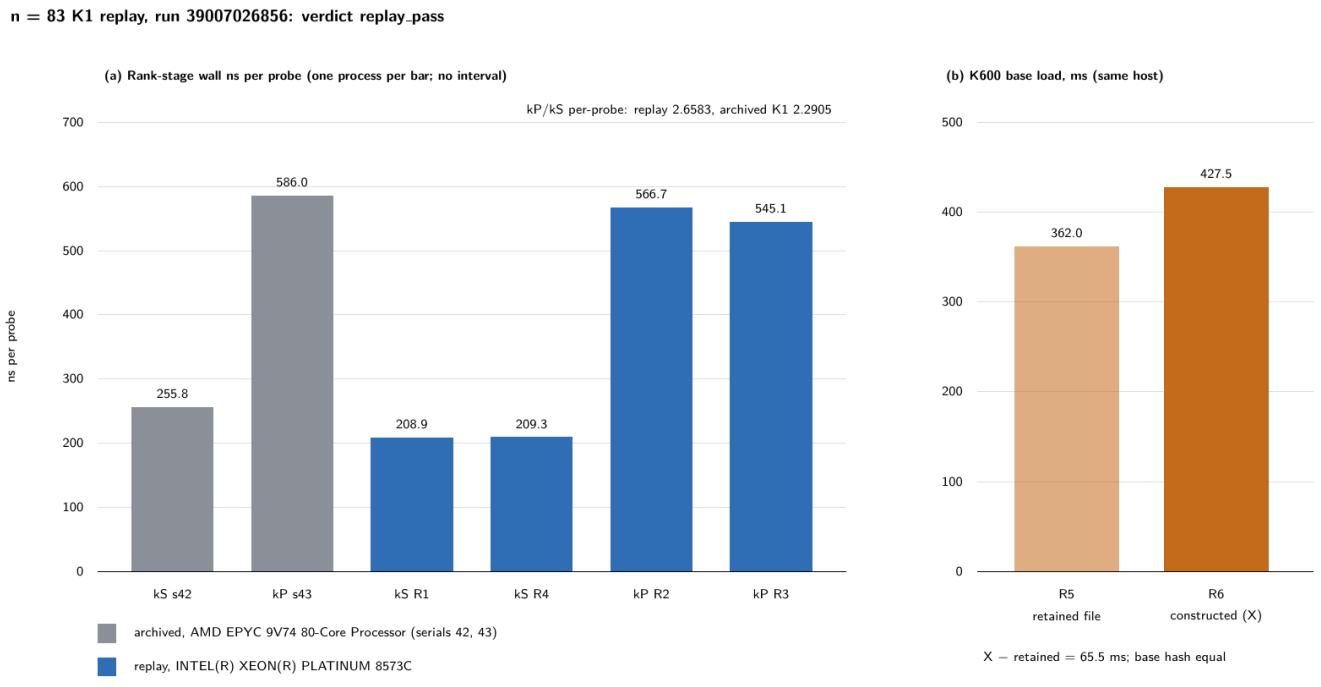


Figure 1: Rank-stage wall time per probe for each rank process on both hosts, and the base load of R5 (retained file) and R6 (constructed), all from the archived records (source, SVG, vector PDF, rendered by `make_figure.py`).

rank run	arm	rank stage (ms)	probes	wall ns per probe
R1	kS	208,961.7	1,000,454,051	208.87
R2	kP	854,793.3	1,508,460,462	566.67
R3	kP	822,191.2	1,508,460,462	545.05
R4	kS	209,430.7	1,000,454,051	209.34
archived serial 42	kS	255,964.7	1,000,454,051	255.85
archived serial 43	kP	884,004.0	1,508,460,462	586.03

per-probe wall ratio kP / kS	replay (Intel 8573C)	archived (EPYC 9V74)
primary: pooled rank stage per probe	**2.6583**	2.2905 (K1); 2.2837 (both blocks, the README's figure)
secondary: first 100 attempts, `rank_attempt_ns`	2.7013	2.2982
all 600 attempts, mean	2.6596	

R3 carries a `major page faults` contention flag (status still `valid`, no issue); it is the faster of the two kP runs, so it does not inflate the ratio. On this host kS is 1.22x faster per probe than on the EPYC (208.9-209.3 against 255.8 ns) and kP 1.05x (545-567 against 586 ns), so the per-probe ratio rose. K1's rank-stage wall ratio is 4.0082 here against 3.4536 in the original run; K1's probe ratio is 1.5078 on both (both blocks: 1.4773).

construction (R5, R6; same host, same job)	value
X = R6 `timing_ms.base_load` (scan x from 1, lift, cofactor projection, first 600 signed-Frobenius orbits, built and indexed)	**427.524 ms**
R5 `base_load` (retained file, parsing only)	361.984 ms
X minus the retained load	**65.541 ms**
`base_scanned_x`	1,266

Interpretation and limits

- **Determinism across host classes: established for K1.** The rank stage's work (every per-attempt probe count, the relation set) is bit-identical on an Intel Xeon 8573C and the original AMD EPYC 9V74, as the protocol required. The host-independent part of the stacked row's gain, the rank-stage operation-count ratio 1.4773 (probes and root calls, both blocks), therefore stands; it is a count fixed by the rank seed.
- **The wall-time headline is host-dependent.** The per-probe wall ratio is 2.66 on this host against 2.28 on the original, so the 3.24x end-to-end wall-time ratio (R = 3.2391 [2.9809, 3.5240]) and every `S` on the row are one-host figures. On this replay host the per-probe factor is larger, which would favour the stack, but the replay re-measured neither the cold arms nor the 20 points, so it gives no new R or `S` on any host (PROTOCOL, "What this does not do"). The row keeps its R and `S` with the host named. One replay on one other host class: n = 2 processes per arm, no interval, no third host.
- **Construction time X: measured, and small.** X = 0.428 s, of which 0.066 s more than loading the retained file. Against the end-to-end cold rows (283.8 s for cS, 919.2 s for cP) it is about 0.15% and 0.05%. With the stacked README's formula (X added to both arms' E2E; `S` = E2E over the run's 29,314.2 ms per unit, rT770 step 316.66 ns x sqrt(r)), the every-phase figures become **R 3.2357** (was 3.2391) and **cold `S` 9.6950** (was 9.6805; gap to the tuned rho 94.56 against 94.42). These are illustrative: they add an Intel-host X to EPYC-host wall times (PROTOCOL: any R(X) built this way mixes two hosts). The K2 / K-N1 caveat (3.2391 an upper bound, 9.6805 a lower bound) is now quantified: the bound is within 0.11% of R and 0.15% of `S` at the measured X. The ratio falls below 2 only at X >= 351.6 s, about 820 times the measured value.
- **Not addressed.** The §5a 0 / 0.1 / 1 GAE band stays uncomputable: the frozen producer emits no group-operation counts, and this replay used the same producer. That needs a producer change under a new protocol (stacked README, follow-up 1).

Files

File	Role
PROTOCOL.md, replay.py	frozen protocol and driver (commit 863f6090)
run_github_39007026856/	the workflow artifact, unzipped byte for byte; ARTIFACT.json provenance, SHA256SUMS
make_figure.py, figures/replay.{tex,svg,pdf,png}	the figure: TikZ source and SVG written from the archived JSON and records; PDF by pdflatex, PNG by pdftoppm
RESULTS.pdf	this report: pandocRESULTS.md-oRESULTS.pdf--pdf-engine=pdflatex-H../koblitz-n83-stacked-20261016/report_header.tex--lua-filter../koblitz-n83-stacked-20261016/report_filter.lua-Vgeometry:margin=2cm-Vfontsize=10pt-Vcolorlinks=true